

E-COMMERCE SALES DASHBOARD USING MICROSOFT POWER BI

INTERNSHIP WORK SUBMITTED IN PARTIAL FULLFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE

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BY

RUBYSHA E (24SBDS043)

GUIDED

BY

Dr. M. Sasikala, M.Sc., M.Phil., Ph.D.

Head of the Department

Department Of Data Science,

PSGR Krishnammal College for Women,

Coimbatore – 641004.

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ABSTRACT

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The rapid growth of e-commerce platforms generates massive volumes of transactional data daily, creating a pressing need for effective business intelligence solutions that can transform raw numbers into meaningful business insights. This report presents the development of an interactive dashboard titled “E-Commerce Sales Dashboard” using Microsoft Power BI, which analyzes a comprehensive e-commerce dataset [34500 e-commerce sales data] containing records of customers, products, categories, regions, orders, revenue, and profitability. The project demonstrates how raw, complex sales data can be processed and converted into a dynamic analytical reporting tool to support data-driven decision-making within the retail and e-commerce sectors.

The primary objective of the project was to transform transactional sales data into a visually appealing and interactive dashboard using Microsoft Power BI Desktop. Power Query was employed for data cleaning, transformation, and preparation, ensuring data integrity. Data Analysis Expressions (DAX) were extensively utilized to build a robust data model and create advanced analytical measures, including performance calculations for revenue, profit margins, average order value, and return rates. The dataset features comprehensive attributes such as Order ID, Customer ID, Product ID, Category, Region, Order Status, Price, and Quantity, allowing for multidimensional analysis.

The completed dashboard is organized into four distinct interactive pages: Executive Sales Overview, Customer Analysis, Product Performance Analysis, and Transactional Drillthrough Details. Key visualizations include KPI cards for top-level monitoring, combination charts for monthly revenue trends, geographic bar charts, a customer leaderboard matrix, and an interactive revenue decomposition tree. Drillthrough capabilities allow users to navigate seamlessly from high-level summaries directly into transaction-level detail.

The dashboard follows a clean and professional design with an optimized user interface. Overall, the E-Commerce Sales Dashboard demonstrates how business intelligence tools can convert large volumes of e-commerce data into actionable insights, helping organizations monitor revenue trends, understand customer behavior, identify top-performing products, and improve strategic planning.

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INTRODUCTION

1: INTRODUCTION

1.1 Organization Profile

Decodet Technologies is a leading Information Technology (IT) services and technology consulting company that focuses on providing innovative software solutions, digital transformation services, and industry-oriented training programs. The organization aims to bridge the gap between academic learning and industrial requirements by offering practical exposure through internships, workshops, and real-time projects.

It provides students and professionals with opportunities to gain hands-on experience in modern technologies, enabling them to develop technical expertise and industry-ready skills.

The company specializes in various domains, including Business Intelligence (BI), Data Analytics, Artificial Intelligence (AI), Machine Learning (ML), Cloud Computing, Web Development, Mobile Application Development, and Software Engineering. Through its project-based learning approach, Decodet Technologies encourages learners to solve real-world business problems using advanced technologies and analytical tools.

During the internship, practical knowledge was gained in Microsoft Power BI, Power Query, Data Analysis Expressions (DAX), data modelling, and dashboard development. The internship provided valuable exposure to the complete Business Intelligence workflow, including data collection, data cleaning, transformation, modelling, visualization, and report generation. The experience also enhanced analytical thinking, problem-solving abilities, and the capability to transform raw business data into meaningful visual insights that support effective decision-making.

Overall, the internship at Decodet Technologies provided an excellent platform to understand industry practices, improve technical proficiency, and apply Business Intelligence concepts in a practical environment.

1.2 Objectives of the Internship Project

The primary objective of the E-Commerce Sales Dashboard project is to develop an interactive Business Intelligence solution that converts raw e-commerce transaction data into meaningful business insights using Microsoft Power BI. The project focuses on applying Business Intelligence.

The specific objectives of the project are as follows:

To analyze e-commerce sales transactions and identify meaningful business insights that support strategic decision-making.

To prepare and transform raw transactional data using Power Query to ensure accuracy and data quality.

- To build a robust Power BI data model utilizing dimension and fact tables.
- To develop a comprehensive and interactive E-Commerce Sales Dashboard using Microsoft Power BI Desktop.
- To create Key Performance Indicators (KPIs) such as Total Revenue, Total Profit, Total Orders, and Average Order Value for monitoring executive business performance.
- To implement Data Analysis Expressions (DAX) for creating measures and calculated columns to support analytical reporting.
- To analyze customer behavior by examining age groups, gender distribution, and identifying high-value customers.
- To evaluate product performance by analyzing category contributions, profit margins, and return rates.
- To analyze regional sales performance and understand geographic revenue concentration.
- To implement interactive filtering through slicers and advanced drillthrough functionality for transaction-level analysis.
- To build a professional Business Intelligence dashboard that presents clear, actionable business insights.

1.3 Scope of the Project

The scope of the E-Commerce Sales Dashboard project is to develop an interactive Business Intelligence solution using Microsoft Power BI for analysing digital retail data. The project focuses on transforming raw order data into meaningful business insights through data modelling, DAX calculations, KPI cards, interactive visualizations, and drillthrough analysis.

The dashboard covers four major areas: Executive Sales Overview, Customer Analysis, Product Performance Analysis, and Transactional Drillthrough. It includes important indicators such as Revenue, Profit, Total Orders, Customer Base, Quantity Sold, and Return Rate, enabling users to assess overall business performance efficiently.

The project also uses interactive visualizations such as trend charts, category analysis, scatter plots, matrix tables, and decomposition trees. Interactive filtering and drillthrough functionality allow users to move from summarized business insights to detailed transaction-level information. Overall, the project demonstrates how Power BI can transform e-commerce data into an interactive and centralized analytical solution for data-driven decision-making

PROJECT DESCRIPTION

2. PROJECT DESCRIPTION

2.1 Introduction

The E-Commerce Sales Dashboard is a Business Intelligence (BI) project developed using Microsoft Power BI to analyze, visualize, and interpret online retail sales data in an interactive and meaningful way. E-commerce businesses generate exceptionally large volumes of transactional data daily. Although this raw data contains valuable business information, it is heavily structured in uninterpretable tabular formats, making it difficult to identify revenue patterns, product performance, and customer trends without specialized analytical tools.

Business Intelligence solutions play an important role in converting this raw transactional data into actionable insights. Microsoft Power BI provides powerful capabilities for data integration, cleaning, modelling, visualization, and reporting, enabling organizations to monitor digital storefront performance effectively. This project demonstrates how Power BI can be used to transform large-scale e-commerce datasets into an interactive application that supports informed decision-making.

The dashboard is designed to provide a comprehensive analysis across multiple facets of an e-commerce business, including revenue, profit, customers, products, orders, and regions. Interactive elements such as KPI cards, matrices, scatter plots, and decomposition trees were incorporated to make the analysis intuitive.

The project followed a systematic BI workflow beginning with data import, followed by data cleaning and transformation using Power Query. The cleaned data was then optimized through data modelling to establish relationships between fact and dimension tables. DAX (Data Analysis Expressions) was utilized to create advanced calculated measures. Finally, the processed data was presented across four focused dashboard pages, enabling users to explore the dataset dynamically and derive immediate business value.

2.2 Problem Statement

E-commerce businesses process a continuous stream of sales transactions every day across various regions, product categories, and customer demographics. These transactions include complex information related to pricing, discounts, shipping, returns, and customer details. As business operations scale, managing and extracting value from such large datasets becomes increasingly difficult.

In many retail organizations, raw transactional data is difficult to interpret manually and often fails to provide the immediate insights required for agile business planning. Decision-makers struggle with monitoring overall revenue and profitability, comparing the performance of diverse product categories, identifying high-value customers, and tracking regional operational performance. Furthermore, identifying order behavior, understanding product returns, and monitoring delivery efficiencies are time-consuming when relying on static or manual reporting methods.

Managers and executives require a centralized, interactive reporting environment that allows them to filter data dynamically, compare business categories, and drill down into granular transaction details.

2.3 Tools and Technologies Used

The successful development of the E-Commerce Sales Dashboard required the use of several Business Intelligence tools and technologies. Each tool played an integral role across the data collection, preprocessing, modelling, analysis, and visualization phases.

Microsoft Power BI Desktop

Microsoft Power BI Desktop served as the primary software platform for developing the dashboard. It provided the unified environment needed for importing data, defining relationships, writing analytical measures, designing visual layouts, and configuring interactive reports.

Power Query

Power Query is a data transformation and preparation engine integrated into Power BI. During this project, Power Query was utilized to inspect the raw e-commerce dataset, correct data types, handle any inconsistencies, standardize text formats, and ensure date fields and numerical identifiers were accurately prepared before loading them into the data model.

Data Analysis Expressions (DAX)

Data Analysis Expressions (DAX) is the formula language native to Power BI, essential for creating custom measures and calculated columns. DAX was extensively used to compute Key Performance Indicators (KPIs) such as Total Revenue, Total Profit, Average Order Value, Profit Margin %, and advanced time-intelligence metrics like Revenue MoM %. A calculated column for Age Group was also developed using DAX to categorize customers for demographic analysis.

Data Modelling

Effective data modelling is vital for accurate reporting in Power BI. The project utilized a structured model comprising dimension tables such as Dim_Customer, Dim_Product, Dim_Location, and Dim_Calendar, together with a central fact table, Fact_Sales. This schema supports optimized querying and enables seamless cross-filtering across the dashboard.

Dataset / CSV

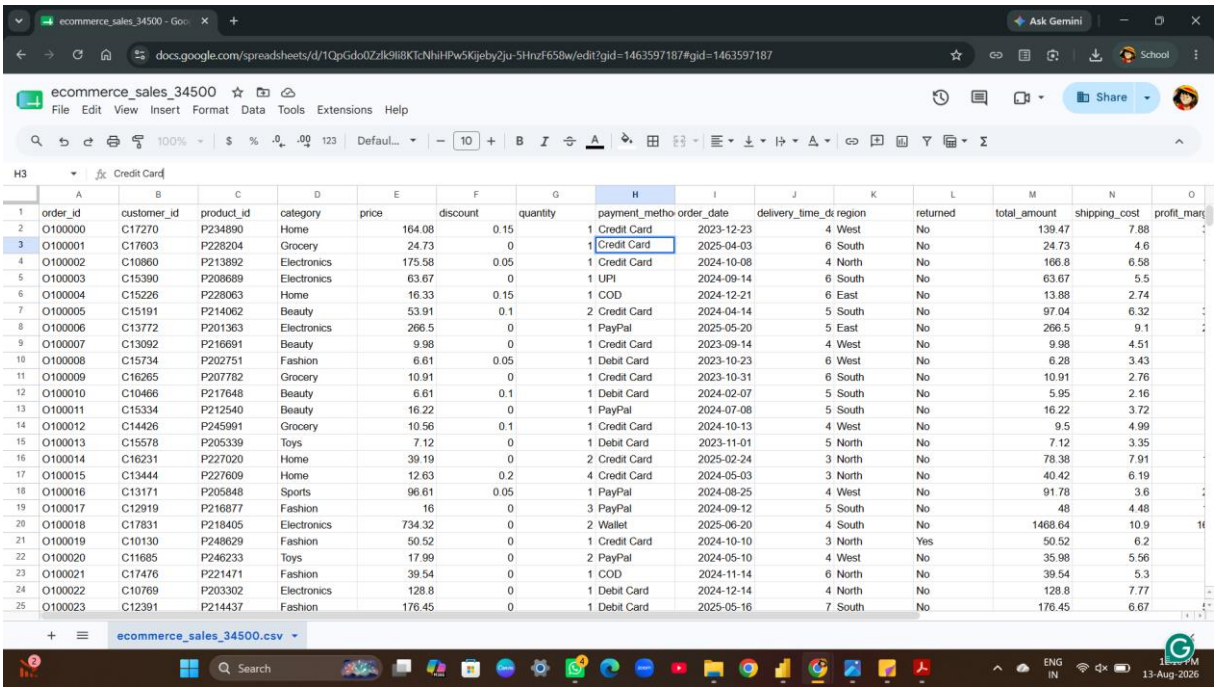
The foundational data source was a detailed e-commerce dataset provided in a structured flat-file format, containing transaction-level records of digital sales.

DATASET DESCRIPTION

3. DATASET DESCRIPTION

3.1 Data Source

The dataset used for this project is an E-Commerce Sales Dataset containing detailed records of online retail transactions. The raw data was prepared and directly imported into Microsoft Power BI Desktop to serve as the foundation for visualization and analysis. It comprises granular data capturing the end-to-end details of customer orders, encompassing the product purchased, financial metrics, customer demographics, and delivery information.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	order_id	customer_id	product_id	category	price	discount	quantity	payment_metho	order_date	delivery_time_d	region	returned	total_amount	shipping_cost	profit_marg
2	O100000	C17270	P234890	Home	164.08	0.15	1	Credit Card	2023-12-23	4	West	No	139.47	7.88	
3	O100001	C17603	P228204	Grocery	24.73	0	1	Credit Card	2025-04-03	6	South	No	24.73	4.6	
4	O100002	C10860	P213892	Electronics	175.58	0.05	1	Credit Card	2024-10-08	4	North	No	166.8	6.58	
5	O100003	C15390	P208689	Electronics	63.67	0	1	UPI	2024-09-14	6	South	No	63.67	5.5	
6	O100004	C15226	P228063	Home	16.33	0.15	1	COD	2024-12-21	6	East	No	13.88	2.74	
7	O100005	C15191	P214062	Beauty	53.91	0.1	2	Credit Card	2024-04-14	5	South	No	97.04	6.32	
8	O100006	C13772	P201363	Electronics	266.5	0	1	PayPal	2025-05-20	5	East	No	266.5	9.1	
9	O100007	C13092	P216691	Beauty	9.98	0	1	Credit Card	2023-09-14	4	West	No	9.98	4.51	
10	O100008	C15734	P202751	Fashion	6.61	0.05	1	Debit Card	2023-10-23	6	West	No	6.28	3.43	
11	O100009	C16265	P207782	Grocery	10.91	0	1	Credit Card	2023-10-31	6	South	No	10.91	2.76	
12	O100010	C10466	P217648	Beauty	6.61	0.1	1	Debit Card	2024-02-07	5	South	No	5.95	2.16	
13	O100011	C15334	P212540	Beauty	16.22	0	1	PayPal	2024-07-08	5	South	No	16.22	3.72	
14	O100012	C14426	P245991	Grocery	10.56	0.1	1	Credit Card	2024-10-13	4	West	No	9.5	4.99	
15	O100013	C15578	P205339	Toys	7.12	0	1	Debit Card	2023-11-01	5	North	No	7.12	3.35	
16	O100014	C16231	P227020	Home	39.19	0	2	Credit Card	2025-02-24	3	North	No	78.38	7.91	
17	O100015	C13444	P227609	Home	12.63	0.2	4	Credit Card	2024-05-03	3	North	No	40.42	6.19	
18	O100016	C13171	P205848	Sports	96.61	0.05	1	PayPal	2024-08-25	4	West	No	91.78	3.6	
19	O100017	C12919	P216877	Fashion	16	0	3	PayPal	2024-09-12	5	South	No	48	4.48	
20	O100018	C17831	P218405	Electronics	734.32	0	2	Wallet	2025-06-20	4	South	No	1468.64	10.9	
21	O100019	C10130	P248629	Fashion	50.52	0	1	Credit Card	2024-10-10	3	North	Yes	50.52	6.2	
22	O100020	C11685	P246233	Toys	17.99	0	2	PayPal	2024-05-10	4	West	No	35.98	5.56	
23	O100021	C17476	P221471	Fashion	39.54	0	1	COD	2024-11-14	6	North	No	39.54	5.3	
24	O100022	C10769	P203302	Electronics	128.8	0	1	Debit Card	2024-12-14	4	North	No	128.8	7.77	
25	O100023	C12391	P214437	Fashion	176.45	0	1	Debit Card	2025-05-16	7	South	No	176.45	6.67	

Figure 3.1: E-Commerce Sales Dataset Imported into Microsoft Power BI

The records represent individual line items of e-commerce orders, providing a complete view of digital retail operations. Through the Power BI import process, the raw file was loaded into the Power Query Editor, where it was subsequently structured to support advanced Business Intelligence analysis.

3.2 Dataset Features

The E-Commerce Sales Dashboard is based on a structured data model consisting of a fact table, dimension tables, a calendar table, and analytical measures. The major features used in the dashboard are described below.

Fact_Sales — Transaction Features

- **Order ID:** A unique identifier representing a specific customer order transaction.
- **Date:** The date and timestamp associated with the order transaction, used for chronological and time-based analysis.
- **Customer ID:** A relational identifier linking each transaction to the corresponding customer profile.
- **Product ID:** A relational identifier linking each transaction to the corresponding product or SKU.
- **Price:** The item-level price associated with the transaction.
- **Quantity:** The number of units purchased in a transaction.
- **Revenue:** The gross monetary sales value generated by the transaction.
- **Profit:** The net financial gain generated after accounting for the relevant costs.
- **Delivery Days:** The number of days taken to fulfill and deliver the order.
- **Status:** The operational status of the transaction, including categories such as returned orders.

Dim_Customer — Customer Features

- **Customer ID:** The unique identifier of the customer.
- **Customer Gender:** The gender classification of the customer, including Female, Male, and Other.
- **Customer Age:** The numerical age of the customer.
- **Age Group:** A calculated demographic segmentation based on customer age. The groups are Under 25, 25–34, 35–44, 45–54, 55–64, and 65+.

Dim_Product — Product Features

- **Product ID:** The unique identifier assigned to a product or inventory SKU.
- **Category:** The product classification used to analyze revenue and profitability across product groups.
- **Price:** The base catalog price associated with the product.

Dim_Location — Geographic Features

- **Location ID:** A unique identifier for the geographic market location.
- **Region:** The broader geographic territory in which the transaction occurred, including South, North, West, East, and Central.
- **City:** The city-level geographic attribute associated with the transaction location.

Dim_Calendar — Time Features

- **Date:** The primary date field used to establish the relationship between transactions and the calendar dimension.
- **MonthYear:** A combined month and year field used for monthly trend visualizations.
- **Year:** The calendar year used for time-based filtering and analysis.
- **DayName:** The name of the day of the week, from Monday to Sunday, used for chronological analysis.

Analytical Measures

In addition to the source columns and calculated fields, several DAX measures were created to support the dashboard's KPI cards and analytical visualizations.

- Total Revenue: Calculates the aggregate revenue generated across transactions.
- Total Profit: Calculates the aggregate profit generated across transactions.
- Total Orders: Counts the distinct number of orders.
- Total Quantity Sold: Calculates the total number of units sold.
- Profit Margin %: Measures profitability as a percentage of total revenue.
- Average Order Value: Calculates the average revenue generated per order.
- Returned Orders: Counts orders whose status is recorded as returned.

- Return Rate %: Calculates the proportion of returned orders relative to total orders.
- Purchase Frequency: Measures the av 9 ge number of orders associated with each customer.

The dataset contains a structured combination of numerical, categorical, identifier, geographic, and date-related attributes, making it suitable for comprehensive e-commerce sales and business intelligence analysis. The transaction-level data includes numerical attributes such as Price, Quantity, Revenue, Profit, and Delivery Days, while categorical attributes include Product Category, Customer Gender, Order Status, Region, and City. Identifier fields such as Order ID, Customer ID, Product ID, and Location ID establish relationships between transactions and the corresponding customer, product, and geographic dimensions.

The Fact_Sales table contains the primary transaction-level information, including Order ID, Date, Customer ID, Product ID, Price, Quantity, Revenue, Profit, Delivery Days, and Order Status. These features provide the foundation for analyzing sales volume, revenue generation, profitability, order activity, and delivery performance.

The Dim_Customer table contains customer-related attributes such as Customer ID, Customer Gender, and Customer Age. A calculated Age Group feature was also created to segment customers into Under 25, 25–34, 35–44, 45–54, 55–64, and 65+ age categories. These features support demographic analysis and help identify differences in purchasing behaviour across customer groups.

The Dim_Product table contains Product ID, Category, and Price, enabling product and category-level analysis. The category information supports the comparison of revenue and profitability across different product groups, while product identifiers allow individual products to be analyzed within the dashboard.

The Dim_Location table provides geographic information through Location ID, Region, and City. These features enable the dashboard to compare sales performance across different geographic markets and identify regions contributing significantly to overall revenue.

The Dim_Calendar table provides the temporal structure required for time-based analysis. It includes Date, MonthYear, Year, and DayName, which support monthly trends, yearly comparisons, date-based filtering, and chronological analysis of sales performance.

Before visualization and analysis, the data was prepared and structured within the Microsoft Power BI environment. The data model was organized into a fact table and supporting dimension tables, allowing relationships between transactions, customers, products, locations, and calendar dates to be established. This structured model provides a reliable foundation for interactive filtering, aggregation, and analytical calculations.

Several DAX measures were developed to transform the underlying transaction data into meaningful business metrics. These include Total Revenue, Total Profit, Total Orders, Total Quantity Sold, Profit Margin %, Average Order Value, Returned Orders, Return Rate %, and Purchase Frequency. These measures support the KPI cards and analytical visualizations used throughout the dashboard.

The combination of transaction, customer, product, geographic, and temporal features enables the development of various Power BI visualizations, including KPI cards, monthly revenue and profitability charts, category contribution charts, geographic revenue charts, customer analysis visuals, product performance analysis, decomposition analysis, interactive slicers, and transactional drillthrough tables.

Overall, the E-Commerce Sales Dataset provides a comprehensive foundation for Business Intelligence and sales performance analysis. Its combination of transactional, customer, product, geographic, and time-based features enables users to evaluate revenue and profitability, understand customer purchasing behaviour, compare product categories, analyze regional performance, monitor returned orders, and identify important sales trends. The dataset therefore demonstrates the practical application of Microsoft Power BI, data modelling, DAX, and interactive dashboard development in transforming transactional e-commerce data into meaningful business insights and supporting data-driven decision-making.

IMPLEMENTATION AND ANALYSIS

4. IMPLEMENTATION AND ANALYSIS

4.1 Data Cleaning and Transformation

The initial phase of dashboard development required importing the raw e-commerce dataset into Microsoft Power BI Desktop (Figure 4.1.1). Utilizing the Power Query Editor, the data underwent a systematic cleaning and transformation process to guarantee accuracy, consistency, and analytical readiness.

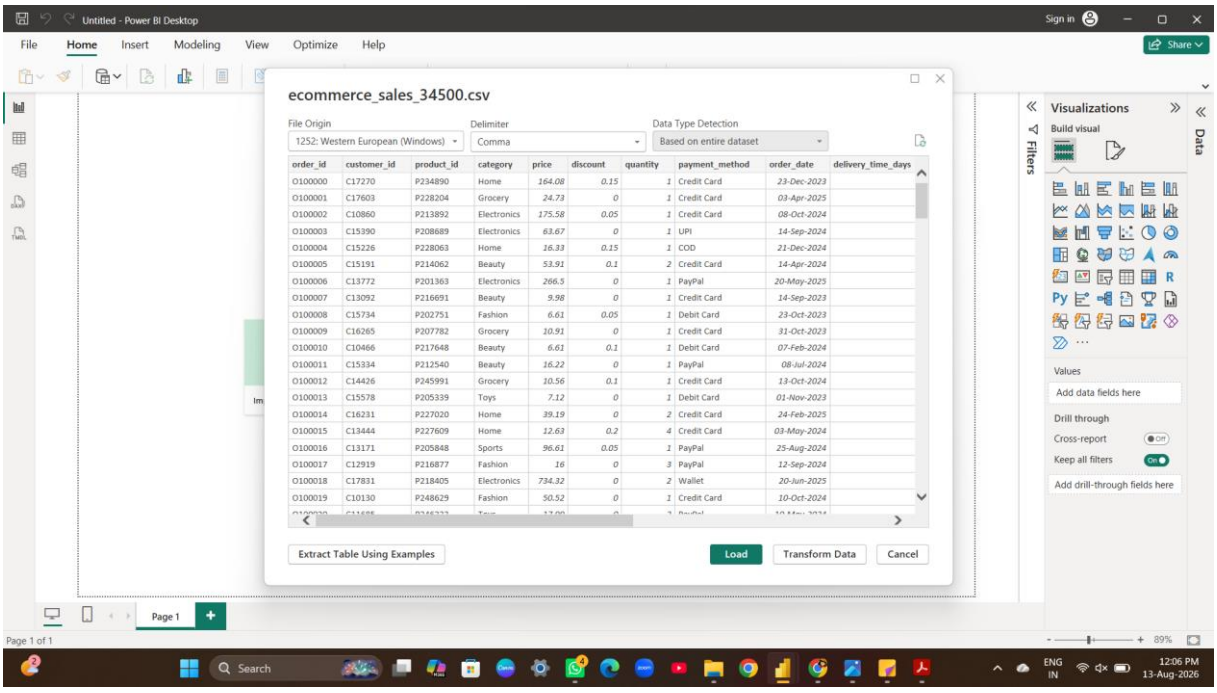


Figure 4.1.1: Importing the E-Commerce Dataset into Microsoft Power BI

During this stage, several essential data preparation steps were executed (Figure 4.1.2). Columns were thoroughly inspected to ensure correct data type classification—verifying that financial fields like Price, Revenue, and Profit were formatted as decimal numbers, and Order Date was properly cast as a Date data type. Identifiers such as Order ID and Customer ID were validated to ensure they functioned correctly as keys for data modelling. The dataset was scanned for missing values, and any inconsistencies or duplicate rows were handled appropriately. Categorical text fields were standardized to avoid duplication in visual filters. These foundational steps ensured the final data model remained robust and reliable for DAX computations.

12

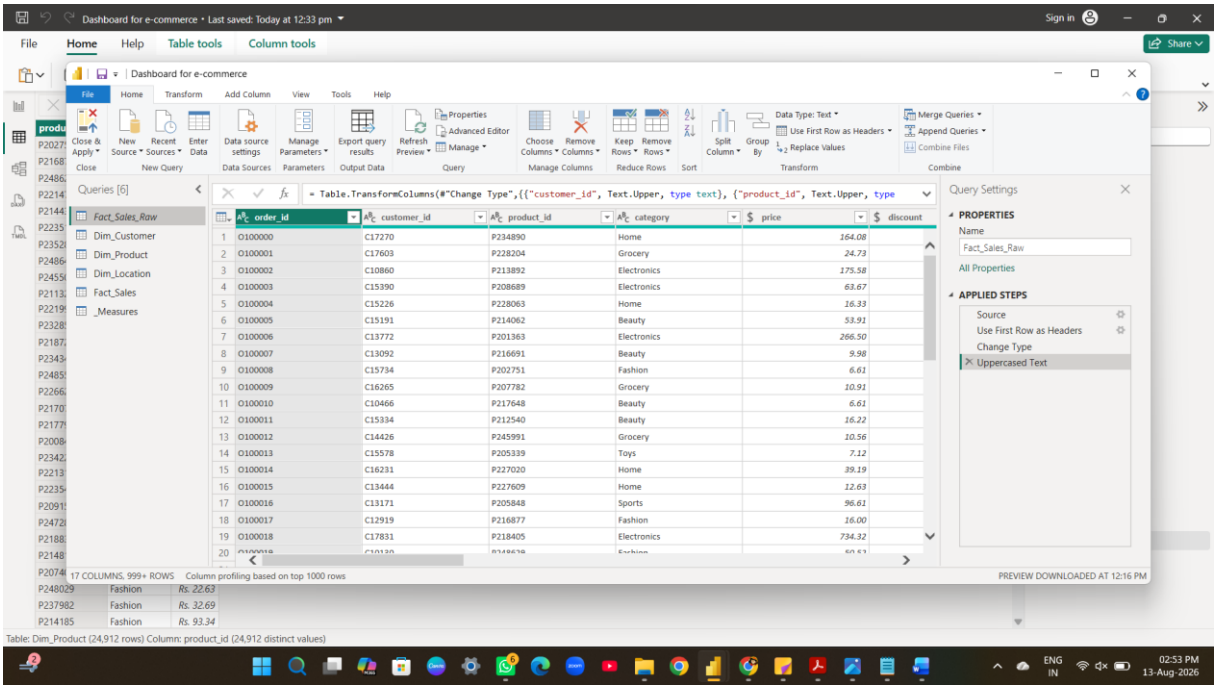


Figure 4.1.2.: Data Cleaning and Transformation of the E-Commerce Dataset

4.2 Data Modelling

Following data transformation, the cleaned tables were loaded into the Power BI Data Model. A logical data architecture was established to optimize calculation performance and enable interactive cross-filtering across visuals.

The model relies on a structured schema integrating a central fact table and supporting dimension tables (Figure 4.2). The core table, Fact_Sales, houses the quantitative transactional metrics such as revenue, profit, and quantity. This is surrounded by dimension tables including Dim_Customer, Dim_Product, Dim_Location, and a dedicated Dim_Calendar table for time-intelligence operations. Relationships were formed using primary and foreign keys, such as linking Customer ID in the fact table to the customer dimension. Proper cardinality and filter directions were established so that filtering by a specific region or date recalculates the metrics in the fact table accurately.

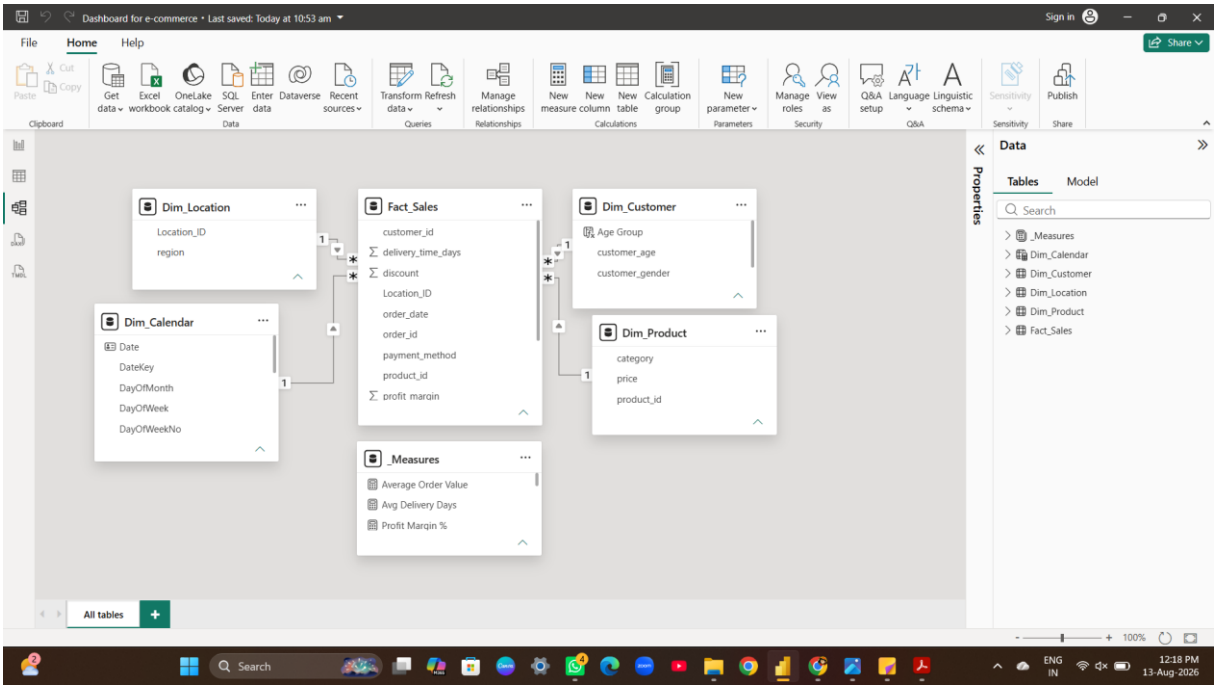


Figure 4.2: Power BI Data Model and Table Relationships for the E-Commerce Sales Dashboard

4.3 Dashboard Overview

The E-Commerce Sales Dashboard was developed in Microsoft Power BI to deliver a comprehensive, user-friendly interface for interpreting complex retail data. The dashboard is structurally divided into four focused report pages (Figure 4.3.1 - Figure 4.3.4).

1. EXECUTIVE SALES OVERVIEW:

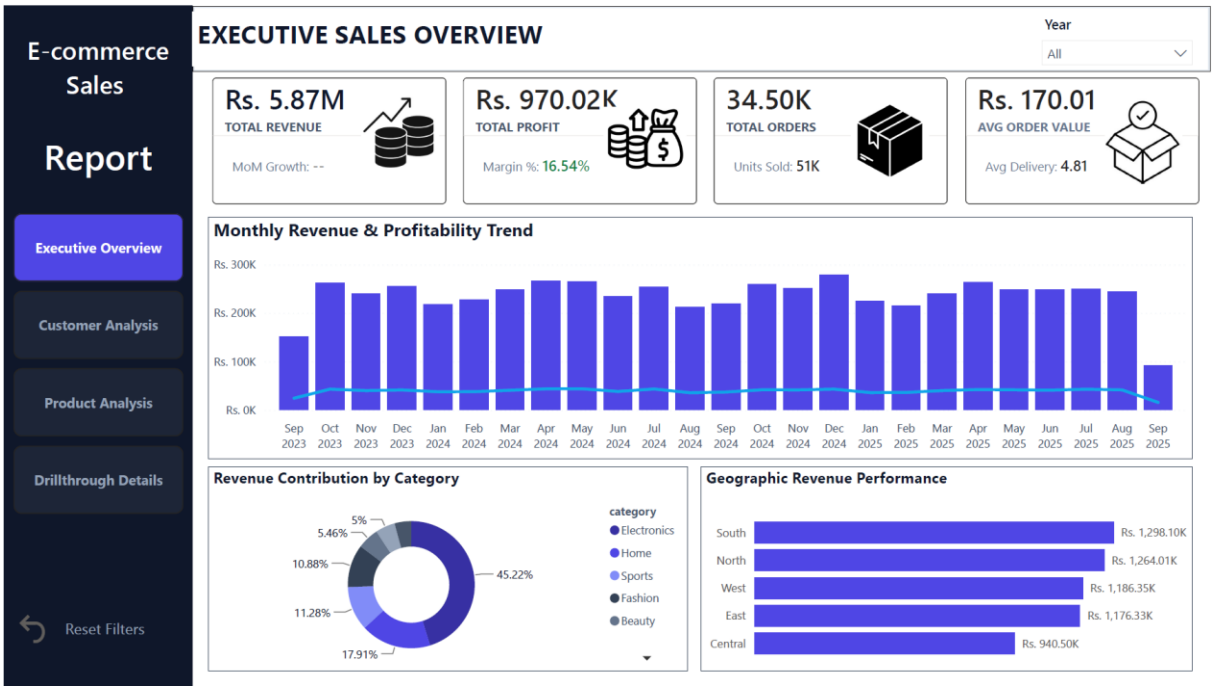


Figure 4.3.1: Executive Sales Overview Dashboard for Monitoring Overall Business Performance

2. CUSTOMER ANALYSIS:

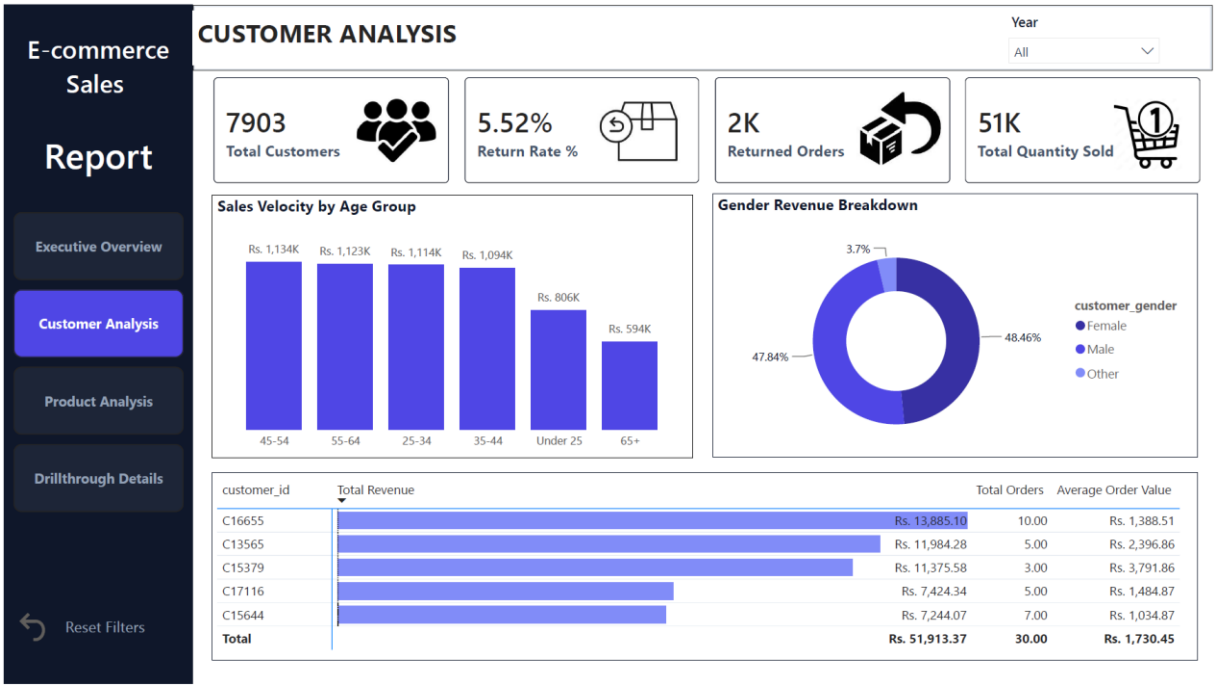


Figure 4.3.2: Customer Analysis Dashboard for Evaluating Customer Behaviour

3. PRODUCT PERFORMANCE ANALYSIS:

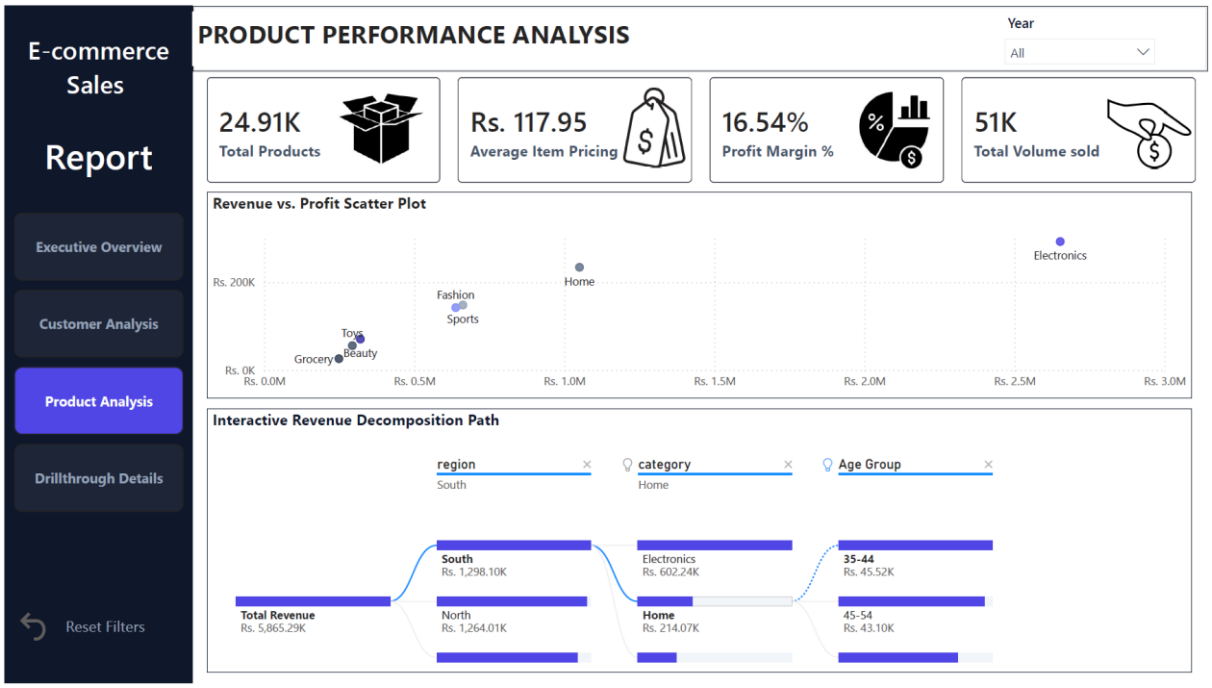
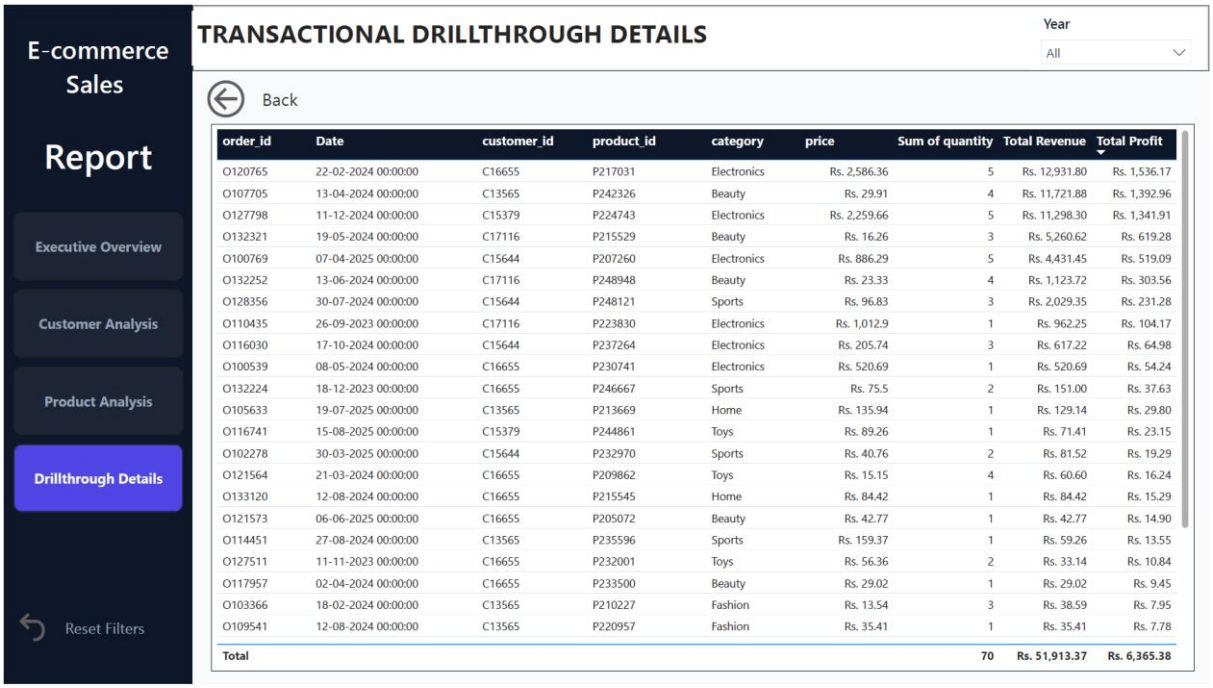


Figure 4.3.3: Product Performance Dashboard for Analysing Product Sales and Profitability

4. TRANSACTIONAL DRILLTHROUGH DETAILS:



order_id	Date	customer_id	product_id	category	price	Sum of quantity	Total Revenue	Total Profit
O120765	22-02-2024 00:00:00	C16655	P217031	Electronics	Rs. 2,586.36	5	Rs. 12,931.80	Rs. 1,536.17
O107705	13-04-2024 00:00:00	C13565	P242326	Beauty	Rs. 29.91	4	Rs. 11,721.88	Rs. 1,392.96
O127798	11-12-2024 00:00:00	C15379	P224743	Electronics	Rs. 2,259.66	5	Rs. 11,298.30	Rs. 1,341.91
O132321	19-05-2024 00:00:00	C17116	P215529	Beauty	Rs. 16.26	3	Rs. 5,260.62	Rs. 619.28
O100769	07-04-2025 00:00:00	C15644	P207260	Electronics	Rs. 886.29	5	Rs. 4,431.45	Rs. 519.09
O132252	13-06-2024 00:00:00	C17116	P248948	Beauty	Rs. 23.33	4	Rs. 1,123.72	Rs. 303.56
O128356	30-07-2024 00:00:00	C15644	P248121	Sports	Rs. 96.83	3	Rs. 2,029.35	Rs. 231.28
O110435	26-09-2023 00:00:00	C17116	P223830	Electronics	Rs. 1,012.9	1	Rs. 962.25	Rs. 104.17
O116030	17-10-2024 00:00:00	C15644	P237264	Electronics	Rs. 205.74	3	Rs. 617.22	Rs. 64.98
O100539	08-05-2024 00:00:00	C16655	P230741	Electronics	Rs. 520.69	1	Rs. 520.69	Rs. 54.24
O132224	18-12-2023 00:00:00	C16655	P246667	Sports	Rs. 75.5	2	Rs. 151.00	Rs. 37.63
O105633	19-07-2025 00:00:00	C13565	P213669	Home	Rs. 135.94	1	Rs. 129.14	Rs. 29.80
O116741	15-08-2025 00:00:00	C15379	P244861	Toys	Rs. 89.26	1	Rs. 71.41	Rs. 23.15
O102278	30-03-2025 00:00:00	C15644	P232970	Sports	Rs. 40.76	2	Rs. 81.52	Rs. 19.29
O121564	21-03-2024 00:00:00	C16655	P209862	Toys	Rs. 15.15	4	Rs. 60.60	Rs. 16.24
O133120	12-08-2024 00:00:00	C16655	P215545	Home	Rs. 84.42	1	Rs. 84.42	Rs. 15.29
O121573	06-06-2025 00:00:00	C16655	P205072	Beauty	Rs. 42.77	1	Rs. 42.77	Rs. 14.90
O114451	27-08-2024 00:00:00	C13565	P235596	Sports	Rs. 159.37	1	Rs. 59.26	Rs. 13.55
O127511	11-11-2023 00:00:00	C16655	P232001	Toys	Rs. 56.36	2	Rs. 33.14	Rs. 10.84
O117957	02-04-2024 00:00:00	C16655	P233500	Beauty	Rs. 29.02	1	Rs. 29.02	Rs. 9.45
O103366	18-02-2024 00:00:00	C13565	P210227	Fashion	Rs. 13.54	3	Rs. 38.59	Rs. 7.95
O109541	12-08-2024 00:00:00	C13565	P220957	Fashion	Rs. 35.41	1	Rs. 35.41	Rs. 7.78
Total						70	Rs. 51,913.37	Rs. 6,365.38

Figure 4.3.4: Transactional Drillthrough Dashboard for Detailed Order-Level Analysis

4.3.1 KPI Cards

The dashboard uses KPI cards to present important business metrics in a simple and easily understandable format (Figure 4.3.1.1).

Executive Sales Overview

- **Total Revenue:** Shows the overall revenue generated from all sales transactions.
- **Total Profit:** Shows the total profit generated from the recorded sales.
- **Total Orders:** Shows the number of unique customer orders.
- **Avg Order Value:** Shows the average revenue generated per order.

Customer Analysis

- **Total Customers:** Shows the number of unique customers.
- **Return Rate %:** Shows the percentage of orders that were returned.
- **Returned Orders:** Shows the total number of returned orders.
- **Total Quantity Sold:** Shows the total number of product units purchased.

Product Performance Analysis

- **Total Products:** Shows the number of distinct products in the product dimension.
- **Average Item Pricing:** Shows the average price of the products.
- **Profit Margin %:** Shows profit as a percentage of total revenue.
- **Total Volume Sold:** Shows the total number of product units sold.

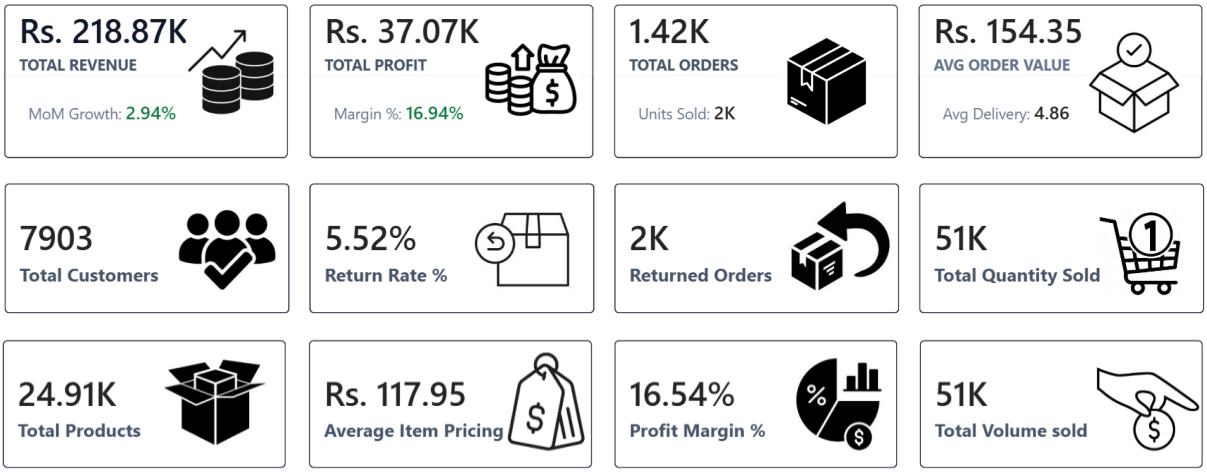


Figure 4.3.1.1: Power BI KPI cards for the E-Commerce Sales Dashboard

4.3.2 Charts and Visualizations

The dashboard incorporates a variety of interactive visualizations to transform the underlying e-commerce data into meaningful business insights. Each visual focuses on a specific analytical perspective, allowing users to understand trends, compare performance, identify patterns, and explore detailed information.

Trend and Distribution Analysis

- **Monthly Revenue & Profitability Trend:** Illustrates how revenue and profit change over time, making it easier to identify growth patterns and fluctuations(Figure 4.3.1.1).
- **Revenue Contribution by Category:** Shows how different product categories contribute to the overall revenue(Figure 4.3.1.2).
- **Geographic Revenue Performance:** Presents revenue across regions to compare market performance(Figure 4.3.1.2).

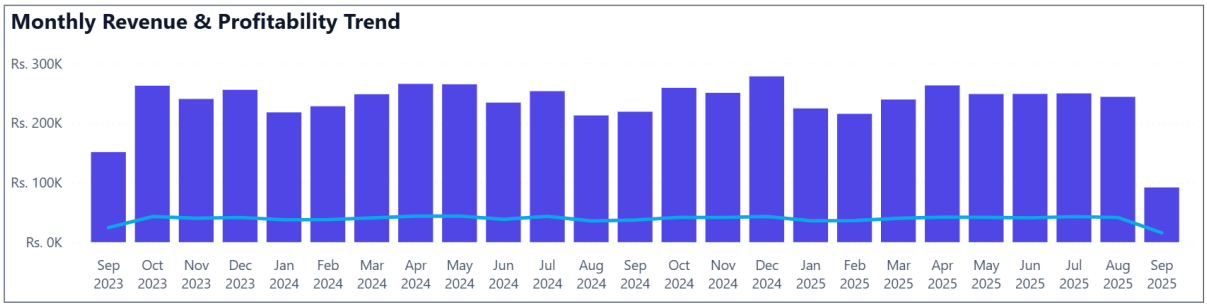


Figure 4.3.1.1: Monthly Revenue and Profitability Trend Using Power BI

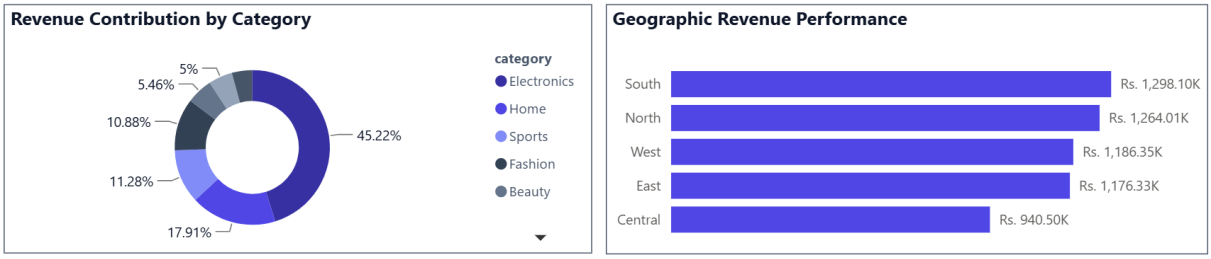


Figure 4.3.1.2: Category-wise Revenue Contribution Analysis Using Power BI

Customer Segmentation

- **Sales Velocity by Age Group:** Compares sales activity among different customer age groups(Figure 4.3.1.3).
- **Gender Revenue Breakdown:** Shows the proportion of revenue generated by different customer gender groups(Figure 4.3.1.3).

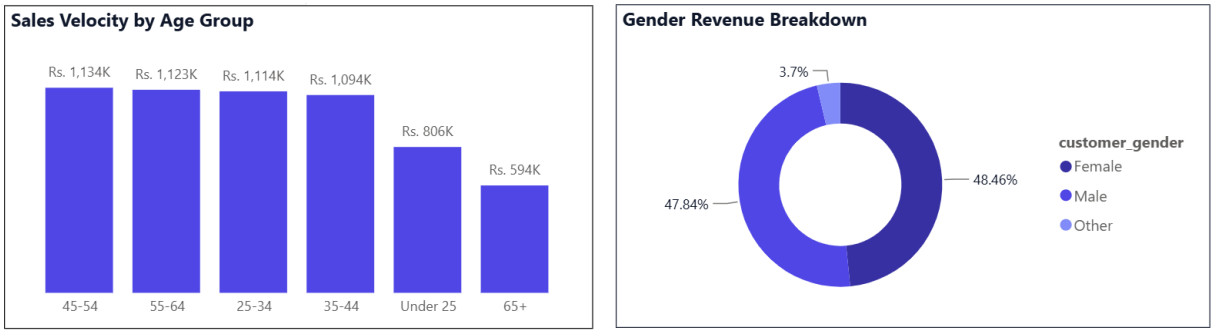


Figure 4.3.1.3: Customer Demographic Revenue and Sales Analysis Using Power BI

Comparative Performance Analysis

- **Matrix Table:** Organizes multiple sales and profitability measures for detailed comparison across business dimensions(Figure 4.3.1.4).
- **Revenue vs. Profit Scatter Plot:** Compares revenue and profit simultaneously to identify relationships and high-performing areas(Figure 4.3.1.5).



Figure 4.3.1.4: Comparative Sales and Profitability Analysis Using Matrix Visualization

customer_id	Total Revenue	Total Orders	Average Order Value
C16655	Rs. 13,885.10	10.00	Rs. 1,388.51
C13565	Rs. 11,984.28	5.00	Rs. 2,396.86
C15379	Rs. 11,375.58	3.00	Rs. 3,791.86
C17116	Rs. 7,424.34	5.00	Rs. 1,484.87
C15644	Rs. 7,244.07	7.00	Rs. 1,034.87
Total	**Rs. 51,913.37**	**30.00**	**Rs. 1,730.45**

Figure 4.3.1.5: Revenue and Profit Relationship Analysis Using Scatter Plot

Exploratory and Transaction Analysis

- **Interactive Revenue Decomposition Path:** Allows users to progressively break down revenue by different dimensions and explore the factors influencing sales(Figure 4.3.1.6).
- **Drillthrough Detail:** Provides detailed transaction-level records for further investigation of individual orders(Figure 4.3.1.7)..

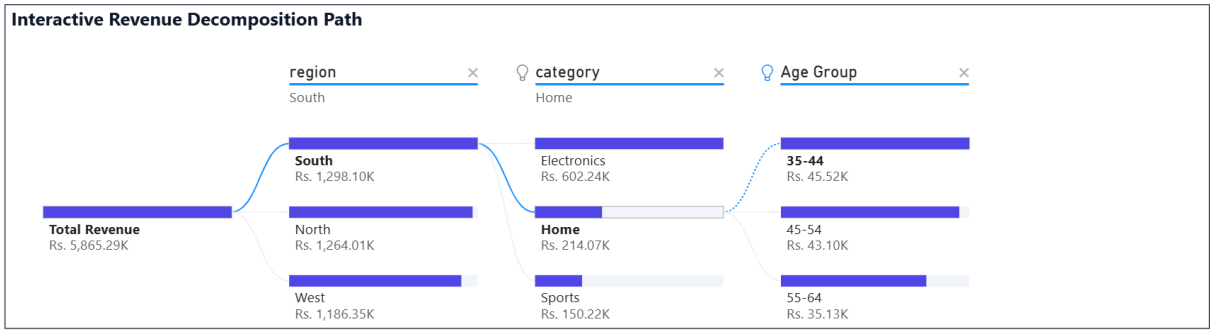


Figure 4.3.1.6: Interactive Revenue Decomposition for Multidimensional Business Analysis

order_id	Date	customer_id	product_id	category	price	Sum of quantity	Total Revenue	Total Profit
O120765	22 February 2024	C16655	P217031	Electronics	Rs. 2,586.36	5	Rs. 12,931.80	Rs. 1,536.17
O107705	13 April 2024	C13565	P242326	Beauty	Rs. 29.91	4	Rs. 11,721.88	Rs. 1,392.96
O127798	11 December 2024	C15379	P224743	Electronics	Rs. 2,259.66	5	Rs. 11,298.30	Rs. 1,341.91
O132321	19 May 2024	C17116	P215529	Beauty	Rs. 16.26	3	Rs. 5,260.62	Rs. 619.28
O100769	07 April 2025	C15644	P207260	Electronics	Rs. 886.29	5	Rs. 4,431.45	Rs. 519.09
O132252	13 June 2024	C17116	P248948	Beauty	Rs. 23.33	4	Rs. 1,123.72	Rs. 303.56
O128356	30 July 2024	C15644	P248121	Sports	Rs. 96.83	3	Rs. 2,029.35	Rs. 231.28
O110435	26 September 2023	C17116	P223830	Electronics	Rs. 1,012.9	1	Rs. 962.25	Rs. 104.17
O116030	17 October 2024	C15644	P237264	Electronics	Rs. 205.74	3	Rs. 617.22	Rs. 64.98
O100539	08 May 2024	C16655	P230741	Electronics	Rs. 520.69	1	Rs. 520.69	Rs. 54.24
O132224	18 December 2023	C16655	P246667	Sports	Rs. 75.5	2	Rs. 151.00	Rs. 37.63
O105633	19 July 2025	C13565	P213669	Home	Rs. 135.94	1	Rs. 129.14	Rs. 29.80
O116741	15 August 2025	C15379	P244861	Toys	Rs. 89.26	1	Rs. 71.41	Rs. 23.15
O102278	30 March 2025	C15644	P232970	Sports	Rs. 40.76	2	Rs. 81.52	Rs. 19.29
O121564	21 March 2024	C16655	P209862	Toys	Rs. 15.15	4	Rs. 60.60	Rs. 16.24
O133120	12 August 2024	C16655	P215545	Home	Rs. 84.42	1	Rs. 84.42	Rs. 15.29
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O127511	11 November 2023	C16655	P232001	Toys	Rs. 56.36	2	Rs. 33.14	Rs. 10.84
O117957	02 April 2024	C16655	P233500	Beauty	Rs. 29.02	1	Rs. 29.02	Rs. 9.45
O103366	18 February 2024	C13565	P210227	Fashion	Rs. 13.54	3	Rs. 38.59	Rs. 7.95
O109541	12 August 2024	C13565	P220957	Fashion	Rs. 35.41	1	Rs. 35.41	Rs. 7.78
Total						**70**	**Rs. 51,913.37**	**Rs. 6,365.38**

Figure 4.3.1.7: Transaction-Level Sales Analysis Using Drillthrough Functionality

CONCLUSION

5. Conclusion

5.1 Conclusion

The E-Commerce Sales Dashboard successfully transforms transactional sales data into a structured and interactive business intelligence solution using Microsoft Power BI. Through data modelling, DAX measures, KPI cards, and interactive visualizations, the dashboard provides a clear view of revenue, profitability, orders, customer behaviour, product performance, regional sales, and transaction details. The use of the Dim_Calendar table and time-intelligence measures further supports effective trend analysis.

The resulting dashboard enables stakeholders to monitor critical operational and financial metrics through four focused analytical areas: Executive Sales Overview, Customer Analysis, Product Performance Analysis, and Transactional Drillthrough. Key metrics such as Total Revenue, Profit Margin, Return Rate, Total Orders, and customer and product indicators are presented in an accessible and dynamically filterable form.

The project also provided practical experience in applying DAX and data modelling techniques to create meaningful analytical measures such as Revenue MoM %, Revenue YoY %, Average Order Value, Purchase Frequency, Return Rate %, and Profit Margin %. These calculations, combined with interactive filtering and drillthrough capabilities, enhance the dashboard's analytical depth and allow users to explore business performance from multiple perspectives.

Overall, the project demonstrates the practical application of Business Intelligence in e-commerce analytics. By converting transactional records into structured measures, interactive visualizations, and drillthrough analysis, the dashboard supports operational understanding, targeted customer analysis, product evaluation, and data-driven decision-making.

5.2 Skills Acquired During Internship

The development of the E-Commerce Sales Dashboard provided practical exposure to Data Analytics and Business Intelligence methodologies.

Technical proficiency was enhanced in Microsoft Power BI Desktop, Power Query, data modelling, and DAX. Hands-on experience with Power Query strengthened skills in data preparation, cleaning, transformation, and validation. Building the relational data model improved understanding of fact-dimension architectures, while DAX measures and calculated columns strengthened analytical calculation and time-intelligence capabilities. Dashboard development also improved understanding of visual hierarchy, chart selection, KPI design, interactive slicers, and drillthrough functionality.

The project also strengthened critical thinking, problem-solving, analytical reasoning, attention to detail, and the ability to translate raw data into meaningful business narratives. These skills contribute to effective communication of analytical findings and the development of practical Business Intelligence solutions.

5.3 Future Scope

While the current E-Commerce Sales Dashboard provides robust analytical capabilities, it establishes a framework that can be expanded significantly.

Future iterations could integrate automated, direct query connections to live e-commerce SQL databases or cloud storage such as AWS or Azure, ensuring that the dashboard reflects more current sales and inventory information rather than relying on static file imports.

The analytical scope could also be broadened through Machine Learning models and Python or R integration for predictive analytics such as revenue forecasting, customer churn prediction, and customer segmentation. Additional supply-chain analytics could further extend the operational value of the dashboard by examining fulfillment and delivery processes in greater detail.

REFERENCES

6. REFERENCES

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2. Microsoft Learn – Power BI Documentation: https://learn.microsoft.com/en-us/power-bi/
3. Microsoft Learn – Power Query: https://learn.microsoft.com/en-us/power-query/
4. Microsoft Learn – DAX: https://learn.microsoft.com/en-us/dax/
5. Microsoft Excel - Official Website https://www.microsoft.com/en-us/microsoft-365/excel
6. Decodet Technologies Internship Training Material & Guidelines (2026).

APPENDIX

7. APPENDIX

7.1 Dashboard Screenshot

The dashboard is structurally divided into four focused report pages (Figure 4.3.1 - Figure 4.3.4)

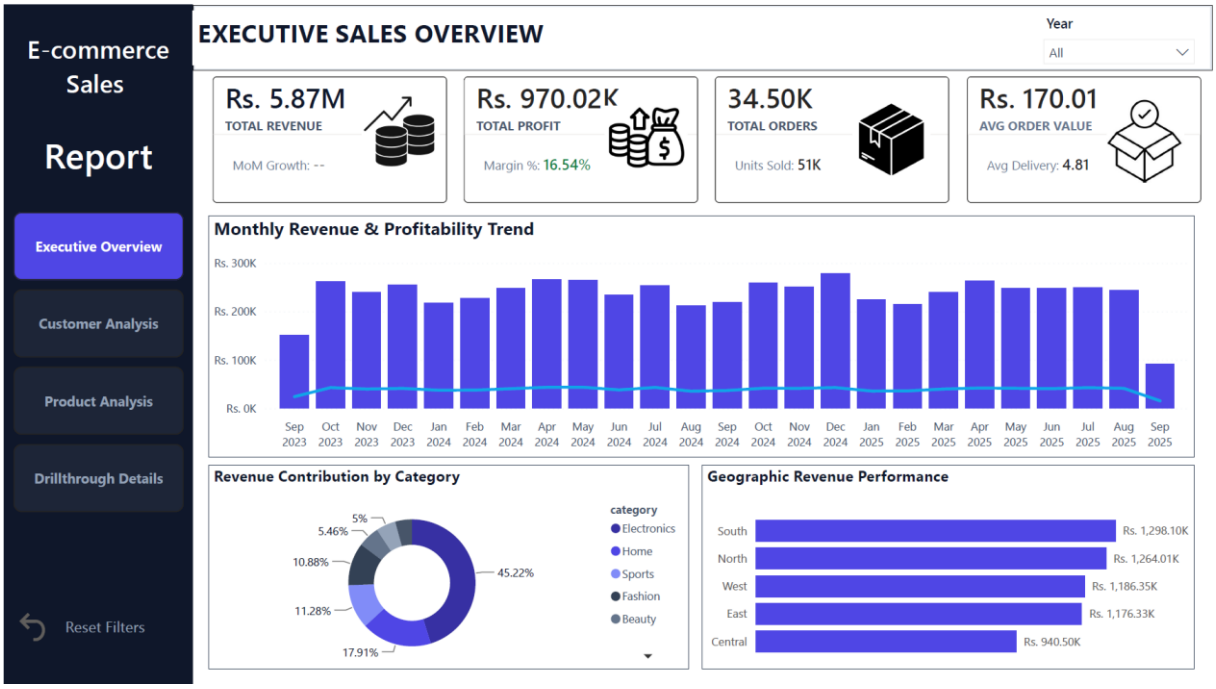


Figure 4.3.1: Executive Sales Overview Dashboard for Monitoring Business Performance

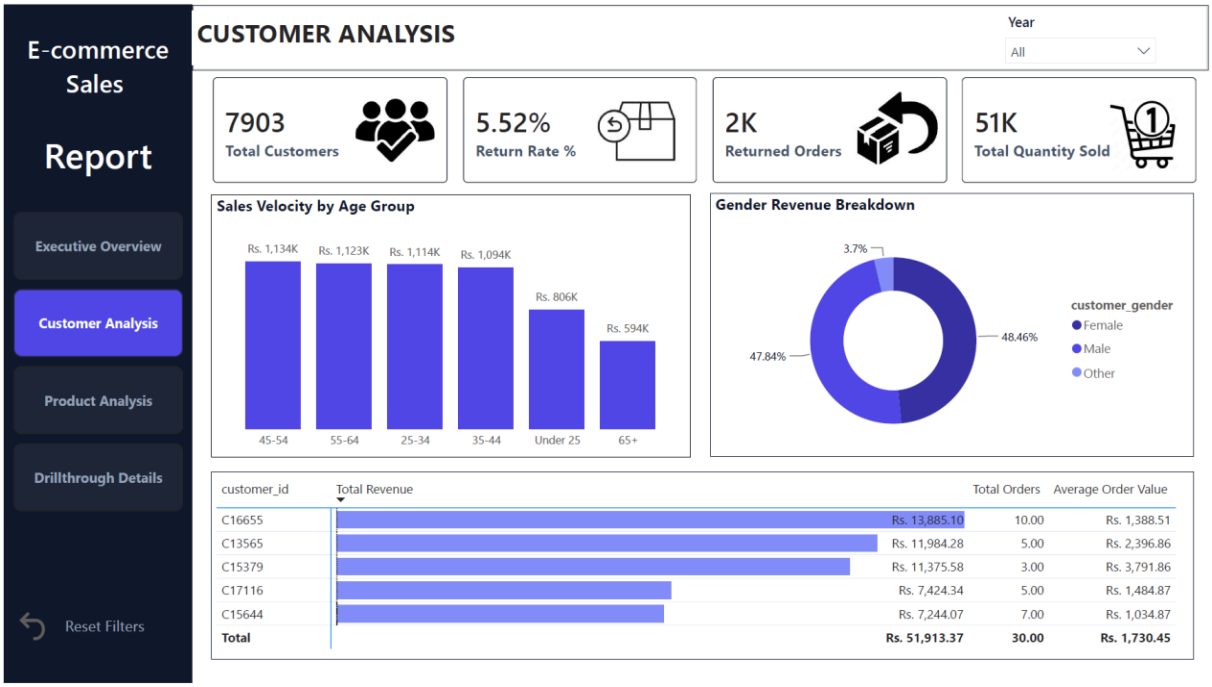


Figure 4.3.2: Customer Analysis Dashboard for Evaluating Customer Behaviour

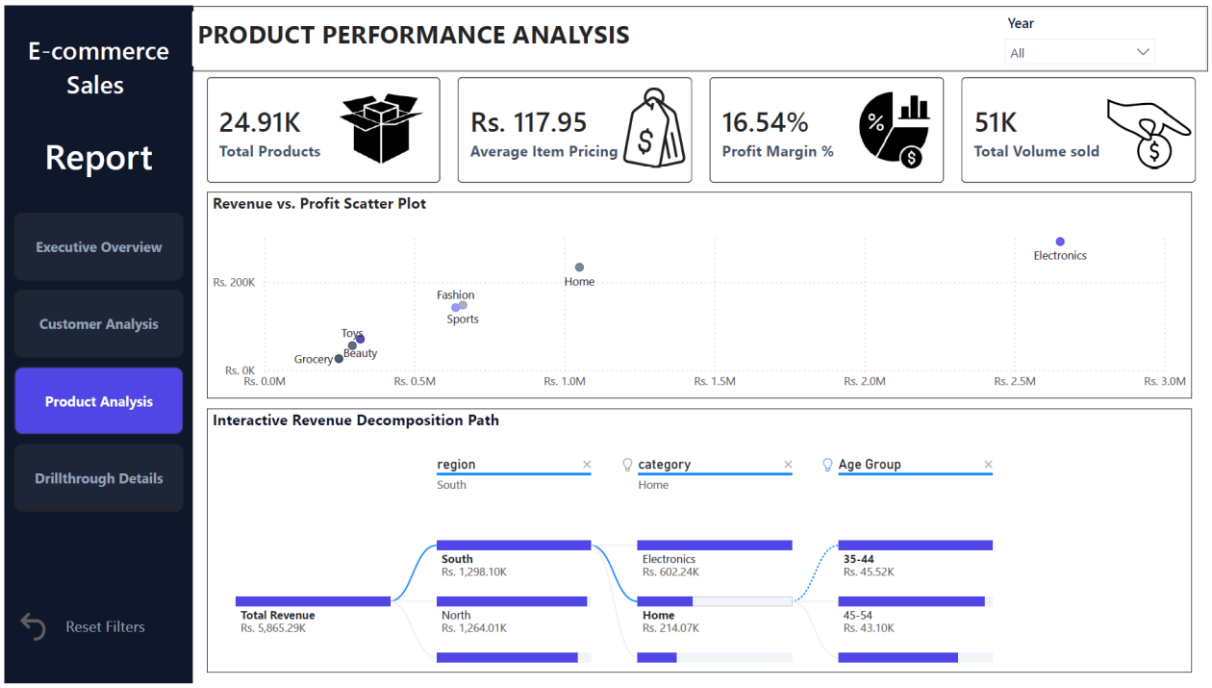
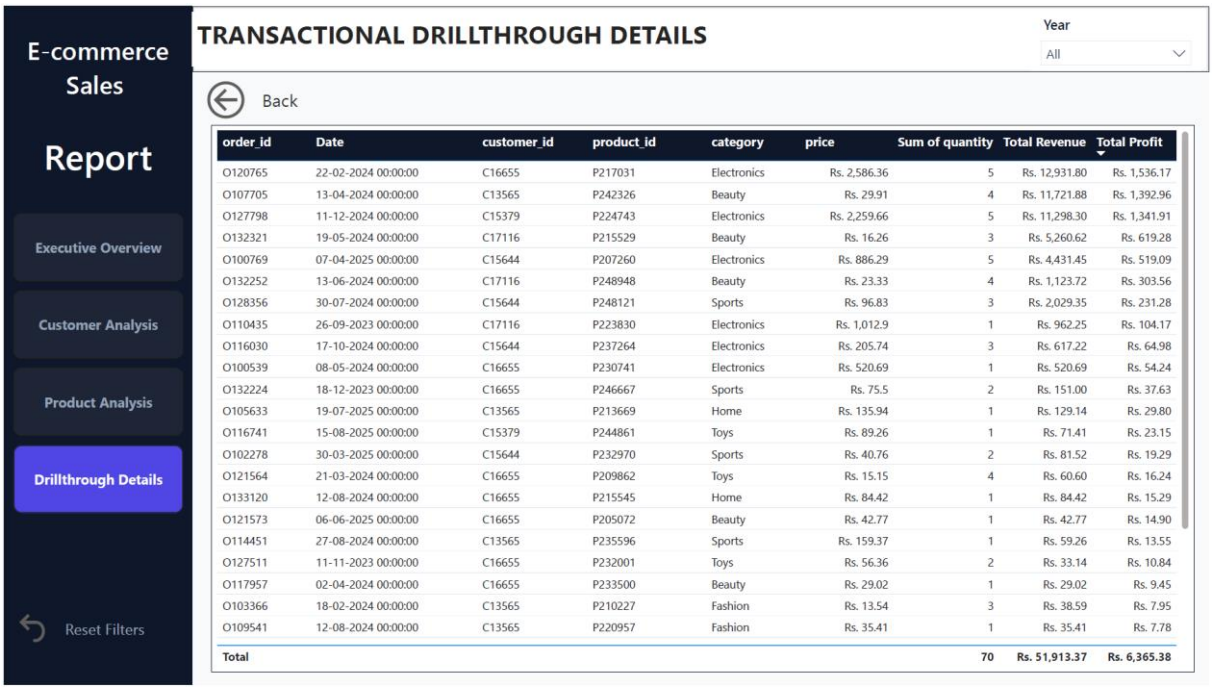


Figure 4.3.3: Product Performance Dashboard for Analysing Product Sales and Profitability



order_id	Date	customer_id	product_id	category	price	Sum of quantity	Total Revenue	Total Profit
O120765	22-02-2024 00:00:00	C16655	P217031	Electronics	Rs. 2,586.36	5	Rs. 12,931.80	Rs. 1,536.17
O107705	13-04-2024 00:00:00	C13565	P242326	Beauty	Rs. 29.91	4	Rs. 11,721.88	Rs. 1,392.96
O127798	11-12-2024 00:00:00	C15379	P224743	Electronics	Rs. 2,259.66	5	Rs. 11,298.30	Rs. 1,341.91
O132321	19-05-2024 00:00:00	C17116	P215529	Beauty	Rs. 16.26	3	Rs. 5,260.62	Rs. 619.28
O100769	07-04-2025 00:00:00	C15644	P207260	Electronics	Rs. 886.29	5	Rs. 4,431.45	Rs. 519.09
O132252	13-06-2024 00:00:00	C17116	P248948	Beauty	Rs. 23.33	4	Rs. 1,123.72	Rs. 303.56
O128356	30-07-2024 00:00:00	C15644	P248121	Sports	Rs. 96.83	3	Rs. 2,029.35	Rs. 231.28
O110435	26-09-2023 00:00:00	C17116	P223830	Electronics	Rs. 1,012.9	1	Rs. 962.25	Rs. 104.17
O116030	17-10-2024 00:00:00	C15644	P237264	Electronics	Rs. 205.74	3	Rs. 617.22	Rs. 64.98
O100539	08-05-2024 00:00:00	C16655	P230741	Electronics	Rs. 520.69	1	Rs. 520.69	Rs. 54.24
O132224	18-12-2023 00:00:00	C16655	P246667	Sports	Rs. 75.5	2	Rs. 151.00	Rs. 37.63
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O116741	15-08-2025 00:00:00	C15379	P244861	Toys	Rs. 89.26	1	Rs. 71.41	Rs. 23.15
O102278	30-03-2025 00:00:00	C15644	P232970	Sports	Rs. 40.76	2	Rs. 81.52	Rs. 19.29
O121564	21-03-2024 00:00:00	C16655	P209862	Toys	Rs. 15.15	4	Rs. 60.60	Rs. 16.24
O133120	12-08-2024 00:00:00	C16655	P215545	Home	Rs. 84.42	1	Rs. 84.42	Rs. 15.29
O121573	06-06-2025 00:00:00	C16655	P205072	Beauty	Rs. 42.77	1	Rs. 42.77	Rs. 14.90
O114451	27-08-2024 00:00:00	C13565	P235596	Sports	Rs. 159.37	1	Rs. 59.26	Rs. 13.55
O127511	11-11-2023 00:00:00	C16655	P232001	Toys	Rs. 56.36	2	Rs. 33.14	Rs. 10.84
O117957	02-04-2024 00:00:00	C16655	P233500	Beauty	Rs. 29.02	1	Rs. 29.02	Rs. 9.45
O103366	18-02-2024 00:00:00	C13565	P210227	Fashion	Rs. 13.54	3	Rs. 38.59	Rs. 7.95
O109541	12-08-2024 00:00:00	C13565	P220957	Fashion	Rs. 35.41	1	Rs. 35.41	Rs. 7.78
Total						70	Rs. 51,913.37	Rs. 6,365.38

Figure 4.3.4: Transactional Drillthrough Dashboard for Detailed Order-Level Analysis

7.2 Data Model Screenshot

The Power BI Model view shows the relationships between the Fact_Sales table and the dimension tables, including Dim_Customer, Dim_Product, Dim_Location, and Dim_Calendar(Figure 7.2.1). These relationships organize the data and enable accurate filtering, aggregation, and analysis across customers, products, regions, and time.

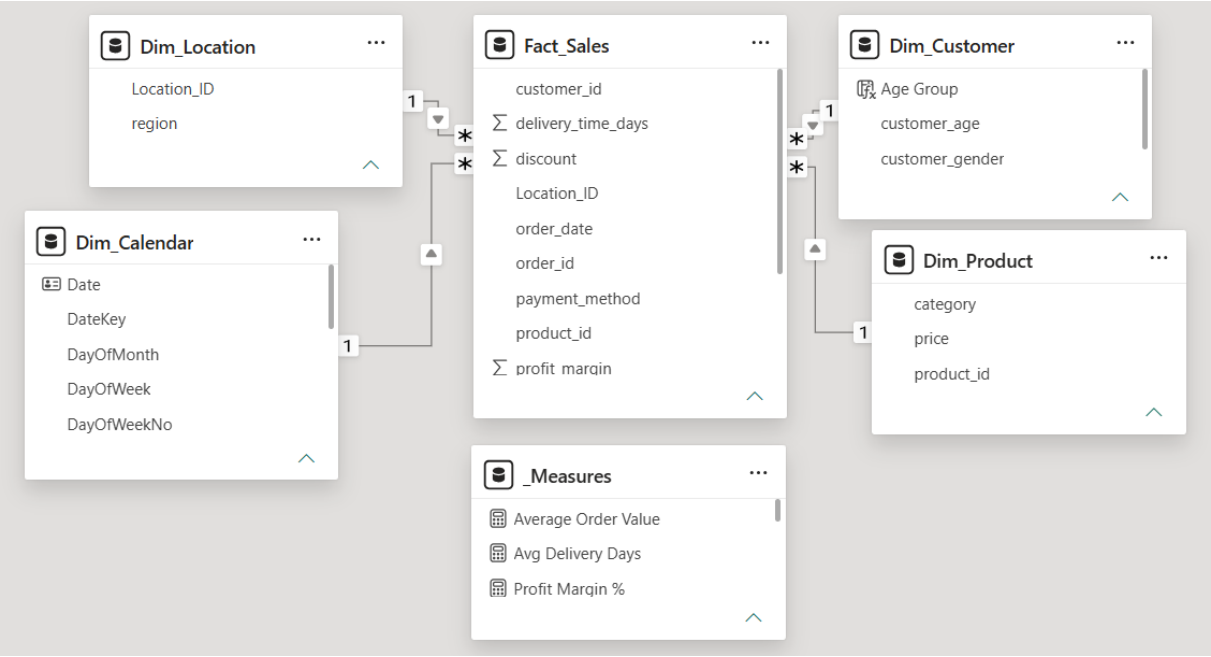


Figure 7.2.1: E-Commerce Data Model and Relationship Structure

Dim_Calendar

The Dim_Calendar table was created using DAX for date-based analysis(Figure 7.2.2). It includes fields such as Date, Month, Quarter, Year, and MonthYear, supporting daily, monthly, quarterly, and yearly analysis. It is linked to Fact_Sales through the Date field and enables time-based filtering, sorting, and MoM and YoY analysis.




```
Dim_Calendar =
VAR _MinYear = YEAR(MIN(Fact_Sales[order_date]))
VAR _MaxYear = YEAR(MAX(Fact_Sales[order_date]))
VAR _StartDate = DATE(_MinYear, 1, 1)
VAR _EndDate = DATE(_MaxYear, 12, 31)

RETURN
ADDCOLUMNS(
    CALENDAR(_StartDate, _EndDate),
    "DateKey", INT(FORMAT([Date], "YYYYMMDD")),
    "Year", YEAR([Date]),
    "QuarterNo", QUARTER([Date]),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "MonthNo", MONTH([Date]),
    "Month", FORMAT([Date], "MMM"),
    "MonthYear", FORMAT([Date], "MMM YYYY"),
    "MonthYearSort", INT(FORMAT([Date], "YYYYMM")),
    "DayOfWeekNo", WEEKDAY([Date], 2), -- 1 = Monday, 7 = Sunday
    "DayOfWeek", FORMAT([Date], "ddd"),
    "DayOfMonth", DAY([Date]),
    "IsWeekend", IF(WEEKDAY([Date], 2) >= 6, "Weekend", "Weekday")
)
```

Figure 7.2.2: DAX Code for Calendar Table Construction Using power BI

7.3 Dax Measures

DAX (Data Analysis Expressions) was used in Microsoft Power BI to create calculated measures and a calculated column for the E-Commerce Sales Dashboard. These calculations were used to generate key performance indicators, analyze customer behaviour, evaluate profitability, monitor returns and delivery performance, and perform time-based revenue analysis.

DAX Measures and Calculated Features

The following DAX measures and calculated features were created in Power BI to support KPI calculations, business analysis, customer segmentation, and time-based analysis.

- **Total Revenue**

  Calculates the total revenue generated from all sales transactions.

```
"Total Revenue = SUM(Fact_Sales[revenue])"
```

- **Total Profit**

  Calculates the total profit generated from the recorded transactions.

```
"Total Profit = SUM(Fact_Sales[profit])"
```

- **Total Orders**

  Calculates the number of unique orders in the dataset.

```
"Total Orders = DISTINCTCOUNT(Fact_Sales[order_id])"
```

- **Total Quantity Sold**

  Calculates the total number of product units sold.

```
"Total Quantity Sold = SUM(Fact_Sales[quantity])"
```

- **Average Order Value**

  Calculates the average revenue generated per order.

```
"Avg Order Value = DIVIDE([Total Revenue], [Total Orders],
0)"
```

- **Profit Margin %**

  Calculates profit as a percentage of total revenue.

```
"Profit Margin % = DIVIDE([Total Profit], [Total Revenue],
0)"
```

- **Returned Orders**

  Calculates the number of orders with a Returned status.

```
"Returned Orders = CALCULATE([Total Orders],
Fact_Sales[status] = "Returned")"
```

- **Return Rate %**

  Calculates the percentage of total orders that were returned.

```
"Return Rate % = DIVIDE([Returned Orders], [Total Orders],
0)"
```

- **Average Delivery Days**

  Calculates the average number of days required to deliver orders.

```
"Avg Delivery Days = AVERAGE(Fact_Sales[delivery_days])"
```

- **Purchase Frequency**

  Calculates the average number of orders placed per customer.

```
"Purchase Frequency = DIVIDE([Total Orders],
DISTINCTCOUNT(Dim_Customer[customer_id]))"
```

- **Customer Age Group – Calculated Column**

  Segments customers into **Under 25, 25–34, 35–44, 45–54, 55–64, and 65+** age groups.

- **Revenue MoM %:**

  Calculates the percentage change in revenue compared with the previous month.

```
"Revenue MoM % = DIVIDE(CurrentMonth - LastMonth,
LastMonth, 0)"
```

- **Revenue YoY %:**

  Calculates the percentage change in revenue compared with the corresponding period of the previous year.

```
"Revenue YoY % = DIVIDE(CurrentRevenue - LastYearRevenue,
LastYearRevenue, 0)"
```